TC-Regularized Learnable Discrete Diffusion

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1

Abstract

We study forward-learned discrete diffusion models with factorized reverse processes and analyze the mismatch induced by statistical dependencies in the forward posterior. We show that this mismatch is characterized by the total correlation of the aggregated forward posterior and propose to explicitly penalize it during training. Empirical results on MNIST illustrate the effect of this regularization under different coupling choices.

1. Introduction

Discrete diffusion models have recently emerged as a powerful alternative to continuous diffusion for modeling discrete or structured data such as text, graphs, and categorical images (Hoogeboom et al., 2021; Austin et al., 2021). A central challenge in this setting is designing diffusion processes that allow for a small number of generative steps while remaining computationally efficient. Recent work has shown that learning the forward process itself can significantly reduce this gap, leading to more efficient discrete diffusion models (Anonymous, 2025).

1.1. Problem Setting

Given a discrete domain $\mathcal{Z}$ with dimensionality D, we consider a discrete diffusion model defined over latent variables $z_t \in \mathcal{Z}^D$ at diffusion time t, with forward process $q(z_t|x)$ and reverse $p_\theta(z_s|z_t)$, where $s = t-1$. For a given forward process, the target distribution for the reverse process is the marginalization of the posterior:

$$q(z_s|z_t) = \int q(x|z_t)q(z_s|z_t,x)dx = \mathbb{E}_{q(x|z_t)}\left[q(z_s|z_t,x)\right] \tag{1}$$

Then, the standard ELBO objective is minimized:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_{q(x)}\mathbb{E}_{t,\,q(z_t|x)}\left[D_{\text{KL}}(q(z_s|z_t,x)\,\|\,p_\theta(z_s|z_t))\right]}_{\mathcal{L}_{\text{cond}}} + \mathbb{E}_{q(x)}\Big[\mathbb{E}_{q(z_0|x)}[-\log p(x|z_0)] + D_{\text{KL}}(q(z_T|x)\,\|\,p(z_T))\Big] \tag{2}$$

1.2. Forward Learned Discrete Diffusion (FLDD)

FLDD (Anonymous, 2025) addresses the mismatch between the target distribution $q(z_s|z_t)$ and a factorized reverse model by learning the forward process itself. In this framework, the reverse process is fixed to be fully factorized:

$$p_\theta(z_s|z_t) = \prod_{i=1}^{D} p_\theta\big(z_s^{(i)}|z_t\big)$$

where $p_\theta\big(z_s^{(i)}|z_t\big) = \text{Cat}\Big(z_s^{(i)};\, v_\theta^{(i)}(z_t,t)\Big)$. To enable a flexible yet tractable forward, the joint distribution is rewritten in a non-Markovian form:

$$q(z_{0:T}|x) = q(z_T|x)\prod_{t=1}^{T} q(z_s|z_t,x). \tag{3}$$

Forward marginals are parameterized to match the factorized structure:

$$q_\phi(z_t|x) = \prod_{i=1}^{D} q_\phi(z_t^i|x) \tag{4}$$

where $q_\phi(z_t^i|x) = \text{Cat}(z_t^i; u_\phi^i(x,t))$, allowing efficient sampling during training. The forward posterior $q_\phi(z_s|z_t,x)$ is required to satisfy the consistency condition

$$q_\phi(z_s|x) = \int q_\phi(z_t|x)\, q_\phi(z_s|z_t,x)\, dz_t, \tag{5}$$

and can be constructed using maximum coupling. Importantly, this modification affects only training, leaving the generative process unchanged.

1.3. Contributions

Building on the FLDD framework, we present a theoretical analysis of the KL-divergence term in the ELBO objective. This analysis serves two purposes: it provides insight

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into why learning the forward process can be beneficial in forward-learned discrete diffusion, and it motivates an additional regularization term that encourages the forward process to induce target distributions that can be effectively matched by a factorized reverse model.

We further propose two practical extensions to FLDD. First, we introduce optimal transport with a learnable cost matrix as an alternative coupling mechanism for constructing the forward posterior. Second, we incorporate variance-reduction baselines for the REINFORCE estimator to improve training stability.

Our implementation can be found here.

2. Methods

2.1. Theoretical Analysis and bounds on the KL divergence

In this section we analyze the following term of our objective, representing the distance between the target posterior $q_\phi(z_s|z_t)$ defined in (1) and the generative process, for a fixed intermediate timestep t:

$$\mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s|z_t)\,\|\,p_\theta(z_s|z_t)\big)\Big] \tag{6}$$

This term can be decomposed as follows (see Appendix A.1 for the full derivation):

$$\begin{aligned}&\mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s|z_t)\,\|\,p_\theta(z_s|z_t)\big)\Big]\\ &=\mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s|z_t)\,\|\,\prod_{i=1}^{D}q_\phi(z_s^i|z_t)\big)\Big]\\ &\quad+\sum_{i=1}^{D}\mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s^i|z_t)\,\|\,p_\theta(z_s^i|z_t)\big)\Big]\end{aligned} \tag{7}$$

Note that the second term is a sum of dimension-wise KL-divergences, and can thus be reduced by learning an appropriate reverse p_θ (being the generative process factorized by design). In contrast, the first term measures the degree of statistical dependence across the coordinates of z_s under the forward posterior. By definition, this term corresponds to the *total correlation* (TC) of the forward posterior, $\mathbb{E}_{q_\phi(z_t)}[\mathrm{TC}(z_s|z_t)]$. Crucially this term cannot be reduced by learning θ, as it depends solely on the structure induced by the forward. This motivates introducing a learnable parameter ϕ in the forward process as proposed in FLDD.

Based on this analysis, we further propose to explicitly penalize the TC of the forward posterior, so as to encourage the forward to produce targets that are closer to being factorized, and thus easier for the factorized reverse process to match. Concretely, we consider the modified objective

$$\mathcal{L}(\phi,\theta)=\mathcal{L}_{\mathrm{ELBO}}(\phi,\theta)+\lambda\,\mathbb{E}_{q_\phi(z_t)}[\mathrm{TC}(z_s|z_t)] \tag{8}$$

where $\lambda\geq 0$ controls the strength of the regularization. Similar ideas have been explored in the context of variational autoencoders to encourage disentangled representations, most notably in FactorVAE (Kim & Mnih, 2019), and β-TCVAE (Chen et al., 2019). However, to the best of our knowledge, this type of regularization has not been previously applied to discrete diffusion models. Table 1 demonstrates the effectiveness of our approach, comparing it to FLDD for different values of λ.

Estimating the total correlation Unfortunately, the total correlation term $\mathrm{TC}(z_s|z_t)$ depends on the intractable aggregated posterior $q_\phi(z_s|z_t)=\mathbb{E}_{q(x)}[q_\phi(z_s|z_t,x)]$ and cannot be computed exactly. Thus, we estimate this quantity using a minibatch-based approximation, following the Minibatch Weighted Sampling (MWS) estimator introduced in (Chen et al., 2019).

Given a minibatch $\{x_n\}_{n=1}^B$, we approximate the aggregated posterior by the empirical mixture $\hat{q}(z_s|z_t)=\frac{1}{B}\sum_{n=1}^B q_\phi(z_s|z_t,x_n)$. The TC is then estimated as

$$\widehat{\mathrm{TC}}=\frac{1}{B}\sum_{n=1}^{B}\left(\log\hat{q}(z_s^{(n)}|z_t)-\sum_{i=1}^{D}\log\hat{q}(z_s^{i,(n)}|z_t)\right)$$

where $z_s^{(n)}\sim q_\phi(z_s|z_t,x_n)$.

Remark 2.1 (Relation to the practical ELBO objective). Note that we based our analysis on the ideal KL defined in (6), which involves the true target $q_\phi(z_s|z_t)$ and differs in general from the standard ELBO in Equation (2). As shown in Appendix A.2, the former term is actually a lowerbound of the tractable conditional ELBO:

$$\mathcal{L}_{\mathrm{cond}}\geq\mathbb{E}_{q(z_t)}\left[D_{\mathrm{KL}}(q_\phi(z_s|z_t)\,\|\,p_\theta(z_s|z_t))\right]. \tag{9}$$

Although replacing the conditional KL with the marginal KL in the ELBO invalidates the upper-bound property on the negative log-likelihood, we argue that our heuristic TC-regularization can still be helpful in practice to encourage the forward process to produce a factorized target. Indeed, even if the conditional posterior $q_\phi(z_s|z_t,x)$ is trivially factorized (as is the case when using maximum coupling or elementwise optimal transport), the target $q_\phi(z_s|z_t)$ may still exhibit non-zero total correlation.

2.2. Optimal Transport Coupling

According to (5), the posterior $q_\phi(z_s|z_t,x)$ must define a valid transport plan for the probability mass across consecutive time-steps. While FLDD uses Maximum Coupling for this purpose, here we investigate an alternative coupling mechanism based on element-wise Optimal Transport (OT).

For each coordinate i, let $u_s=u_\phi^i(x,s)\in\Delta^K$ represent

Table 1. Best FID scores with $T = 4$, baseline $\beta = 0.9$, and learned cost matrix

TC Weight	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	10^{0}	10^{1}	**FLDD**
FID	29.78	24.94	22.31	52.57	58.83	68.62	75.30	40.80

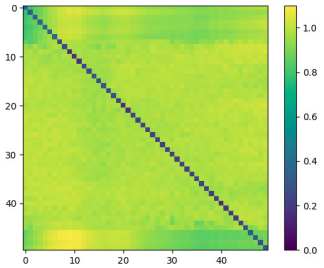

Figure 1. Learned cost matrix for the Sinkhorn transport plan after initialization with $\mathbf{1}_{u \neq v}$ on the mixture-of-two-gaussians dataset, where $\mathcal{Z} = \{0, \ldots, 49\}$ and $D = 2$

our source distribution and $u_t = u_\phi^i(x, t) \in \Delta^K$ the corresponding target distribution. We seek a transport plan $P \in \mathbb{R}_+^{K \times K}$ with $P\mathbf{1} = u_s$, $P^T\mathbf{1} = u_t$. We can then choose a cost matrix $C \in \mathbb{R}^{K \times K}$, where $C_{u,v}$ represents the cost of transporting mass from category u to category v.

We can now solve the following Entropic regularized Optimal Transport problem (Cuturi, 2013):

$$P^* = \arg\min_{P \succeq 0} < P, C > -\varepsilon H(P) \qquad (10)$$
$$\text{s.t.} \quad P\mathbf{1} = u_s \qquad P^T\mathbf{1} = u_t$$

where $H(P) = -\sum_{i,j} P_{ij} \log P_{ij}$ is the entropy and $\varepsilon > 0$. This problem has a unique solution given by the Sinkhorn Scaling and can be found via an iterative algorithm (see Appendix A.5 for the details).

Although the Sinkhorn iterations will slightly increase the computational complexity as compared to maximum coupling, this approach introduces a flexible cost matrix C for the OT problem which we can customize (in our first experiments we fix $C_{u,v} = \mathbf{1}_{u \neq v}$). Furthermore, being this part of the objective now smoothly differentiable, the cost matrix could even be learned for the task at hand. In our experiments, learning the cost matrix consistently yields better results (as demonstrated in Table 2 and Appendix A.6). Figure 1 shows the final cost matrix learned by our algorithm on the so-called mixture-of-two-gaussians dataset.

Table 2. Best FID scores for the different transport plans

Plan	Max. Coupling	OT Fixed	OT Learned
FID	31.55	27.88	23.32

2.3. REINFORCE Variance Reduction

Consider the KL divergence in $\mathcal{L}_{\text{cond}}$. For brevity denote

$$\mathrm{KL}(z_t, x) = D_{\mathrm{KL}}\big(q_\phi(z_s|z_t, x) \,\|\, p_\theta(z_s|z_t)\big).$$

In (Anonymous, 2025), the REINFORCE estimator proposed for $\nabla_\phi \mathcal{L}_{\text{diff}}$ is

$$\hat{g}_\phi = \mathbb{E}_{u(t) q_\phi(z_t|x)} \left[\nabla_\phi \left(\frac{q_\phi(z_t|x)}{[\, q_\phi(z_t|x) \,]_{\text{sg}}} \, \mathrm{KL}(z_t, x) \right) \right] \qquad (11)$$

where $[\,\cdot\,]_{\text{sg}}$ denotes the stop-gradient operator. Although the authors claim that equation (11) is the standard REINFORCE estimator, this is only true if $\mathrm{KL}(z_t, x)$ is independent of ϕ. See Appendix A.3 for details.

In practice $\mathrm{KL}(z_t, x)$ does depend on ϕ through $q_\phi(z_s|z_t, x)$. Applying the product rule to (11) yields two terms:

$$\nabla_\phi \Big(\frac{q_\phi}{[\, q_\phi \,]_{\text{sg}}} \, \mathrm{KL}(z_t, x) \Big)$$
$$= \frac{q_\phi}{[\, q_\phi \,]_{\text{sg}}} \Big(\mathrm{KL}(z_t, x) \, \nabla_\phi \log q_\phi + \nabla_\phi \mathrm{KL}(z_t, x) \Big).$$

The first term has the form of a REINFORCE estimator, the second is just the gradient of KL. Baseline methods can be used in the first term to reduce its variance. It is a standard exercise to derive the optimal baseline:

$$b^*(x, t) = \frac{\mathbb{E}_{q_\phi(z_t|x)} \Big[\|\nabla_\phi \log q_\phi(z_t|x)\|^2 \, \mathrm{KL}(z_t, x) \Big]}{\mathbb{E}_{q_\phi(z_t|x)} \Big[\|\nabla_\phi \log q_\phi(z_t|x)\|^2 \Big]} \qquad (12)$$

See Appendix A.4 for the full derivation of the baseline.

Approximation of baseline A straightforward implementation of the baseline would involve performing Monte Carlo sampling during training to evaluate the expectations with respect to $q_\phi(z_t|x)$. To reduce the computational cost during training and avoid additional sampling, the z_t sampled for training without baseline was used to estimate the expectations in the numerator and denominator. The sample x dependent baseline was approximated by taking the gradient of $\frac{1}{B} \sum_{i=1}^{B} \log q_\phi(z_t|x_i)$, the mean in a given batch of size B, instead of the gradient of $q_\phi(z_t|x)$. The sample dependent KL divergence $\mathrm{KL}(z_t, x)$ is replaced with its mean over the batch size $\mathrm{KL}(z_t) = \frac{1}{B} \sum_{i=1}^{B} \mathrm{KL}(z_t, x_i)$. This means that the approximated baseline would not be sample x dependent and only timestep t dependent. Since no additional sampling is performed, the values v of the approximated numerator and denominator in equation (12) are updated via a moving average $v_{\text{new}} = \beta v_{\text{old}} + (1 - \beta) v_{\text{new}}$, where β is a decay rate. See Table 3 for a comparison between different values of β.

Figure 2. Diffusion steps visualized on the MNIST dataset

Table 3. FID scores with Max. coupling and training with baseline

β	None	0.9	0.99
FID	40.80	32.73	31.55

Figure 3. Final MNIST samples

3. Experiments

On the MNIST dataset, we train the model with 4 timesteps. Grid search was done over the following parameters:

- TC Weight: $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10\}$
- Transport plan: {Maximum Coupling, OT Fixed, OT Learned}
- Approximate baseline in REINFORCE estimator: {None, β = 0.9, β = 0.99}

Evaluation of results is done using the FID metric and we display the mean of the lowest FID scores over 3 runs, to account for randomness. The model was trained for 50 epochs because we found that increasing the number of epochs and the number of warmup and REINFORCE steps as was done in (Anonymous, 2025) did not improve the FID score. 25 epochs were used for warmup and 25 epochs for the REINFORCE step.

In contrast to MNIST, where $\mathcal{Z}$ is merely $\{0, 1\}$ but D is large, we also ran experiments on the mixture-of-two-gaussians dataset from (Anonymous, 2025), where $\mathcal{Z}$ is $\{0, \ldots, 49\}$ but D only 2. We chose to get our dataset from the distribution $0.6 \cdot \mathcal{N}(\mu = (15, 15), \sigma = 5I) + 0.4 \cdot \mathcal{N}(\mu = (25, 35), \sigma = 10I)$. Here two timesteps suffice.

We used Maximum Mean Discrepancy to score the model's output. For both datasets our approach proved superior performance. We provide data from all our experiments in the appendix A.6.

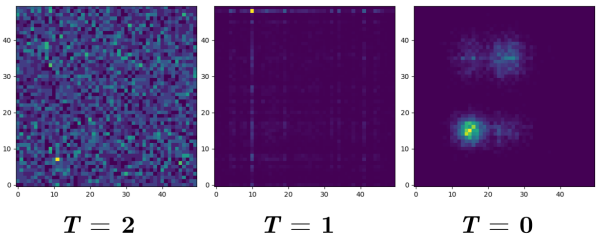

Figure 4. Heatmap of the backwards process at each timestep learned by our model on the mixture-of-two-gaussians dataset

4. Conclusion

Building on the FLDD framework, we showed that the KL-divergence objective decomposes into a total correlation term, which characterizes the mismatch induced by the factorization assumption, and a dimension-wise KL term that can be reduced by learning the reverse process. Crucially, the TC term depends solely on the forward process and cannot be minimized by the reverse model. This analysis motivated our proposed TC-regularization, which explicitly encourages the forward process to induce more factorizable target distributions. Additionally, we introduced optimal transport with a learnable cost matrix as an alternative coupling mechanism and implemented a variance-reduction baseline for the REINFORCE estimator.

Our experiments demonstrate that these extensions yield substantial improvements over the FLDD baseline, reducing FID from 40.80 to 22.31 on MNIST with only four diffusion steps. Each proposed component contributes to this improvement, with the learnable cost matrix and TC-regularization providing complementary benefits. On the mixture-of-two-gaussians task, TC-regularization notably reduced the occurrence of incorrectly mixed samples, where coordinates are drawn from different modes, providing qualitative confirmation of our theoretical analysis.

Future work could explore the impact of these contributions on larger-scale benchmarks and modalities such as text and graph generation, where long-range dependencies may induce more significant mismatches with the factorization assumption. Investigating adaptive schedules for the TC-regularization weight or more expressive cost matrix parameterizations may yield further gains.

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A. Appendix

Lemma A.1 (Projection onto the space of factorized distributions)**.** *Let $q_\phi(z_s \mid z_t)$ be an arbitrary distribution over $z_s = (z_s^1, \dots, z_s^D)$. Consider the set $\mathcal{Q}_{\text{fact}}$ of fully factorized distributions of the form*

$$\tilde{q}(z_s \mid z_t) = \prod_{i=1}^{D} \tilde{q}(z_s^i \mid z_t)$$

Then, the solution of the optimization problem

$$\arg\min_{\tilde{q} \in \mathcal{Q}_{\text{fact}}} \mathrm{KL}(q_\phi(z_s \mid z_t) \,\|\, \tilde{q}(z_s \mid z_t))$$

is given by the product of the marginals of $q_\phi(z_s \mid z_t)$:

$$\tilde{q}^\star(z_s \mid z_t) = \prod_{i=1}^{D} q_\phi(z_s^i \mid z_t)$$

Proof. By definition, the KL divergence can be written as

$$\mathrm{KL}(q_\phi(z_s \mid z_t) \| \tilde{q}(z_s \mid z_t)) = \mathbb{E}_{q_\phi(z_s|z_t)} \left[\log q_\phi(z_s \mid z_t) - \sum_{i=1}^{D} \log \tilde{q}(z_s^i \mid z_t) \right]$$

The first term does not depend on $\tilde{q}$ and can be ignored for the purpose of minimization. Therefore, minimizing the KL divergence is equivalent to maximizing

$$\sum_{i=1}^{D} \mathbb{E}_{q_\phi(z_s|z_t)} \left[\log \tilde{q}(z_s^i \mid z_t) \right]$$

Since the integrand $\log \tilde{q}(z_s^i \mid z_t)$ depends only on the i-th coordinate, the expectation with respect to the joint distribution $q_\phi(z_s \mid z_t)$ reduces to an expectation with respect to the marginal $q_\phi(z_s^i \mid z_t)$:

$$\mathbb{E}_{q_\phi(z_s|z_t)} \left[\log \tilde{q}(z_s^i \mid z_t) \right] = \mathbb{E}_{q_\phi(z_s^i|z_t)} \left[\log \tilde{q}(z_s^i \mid z_t) \right]$$

Thus, the optimization decouples across dimensions, and for each i we must solve

$$\arg\max_{\tilde{q}(z_s^i|z_t)} \mathbb{E}_{q_\phi(z_s^i|z_t)} \left[\log \tilde{q}(z_s^i \mid z_t) \right]$$

subject to $\tilde{q}(z_s^i \mid z_t)$ being a valid probability distribution. The solution of this problem is achieved by setting

$$\tilde{q}(z_s^i \mid z_t) = q_\phi(z_s^i \mid z_t)$$

i.e., matching the true marginal distribution. Combining the optimal solutions for all dimensions yields

$$\tilde{q}^\star(z_s \mid z_t) = \prod_{i=1}^{D} q_\phi(z_s^i \mid z_t)$$

which concludes the proof. □

A.1. ELBO Decomposition

In this section we analyze the following term of our objective, representing the distance between the target posterior $q_\phi(z_s|z_t)$ defined in (1) and the generative process, for a fixed intermediate timestep t:

$$\mathbb{E}_{q_\phi(z_t)} \Big[\mathrm{KL}\big(q_\phi(z_s|z_t) \,\|\, p_\theta(z_s|z_t) \big) \Big] \tag{13}$$

Let $q_\phi^\star(z_s|z_t)$ denote the best factorized approximation of $q_\phi(z_s|z_t)$, defined as the projection of $q_\phi(z_s|z_t)$ onto the space of fully factorized distributions. From Lemma A.1 this is given by the product of the marginal distributions of $q_\phi(z_s|z_t)$:

$$q_\phi^\star(z_s|z_t) = \prod_{i=1}^{D} q_\phi(z_s^i|z_t). \tag{14}$$

Using this : We can now decompose the KL divergence as follows:

$$\begin{aligned}
&\mathrm{KL}\big(q_\phi(z_s|z_t) \,\|\, p_\theta(z_s|z_t)\big) \\
&= \mathbb{E}_{q_\phi(z_s|z_t)}\left[\log \frac{q_\phi(z_s|z_t)}{q^\star_\phi(z_s|z_t)}\right] + \mathbb{E}_{q_\phi(z_s|z_t)}\left[\log \frac{q^\star_\phi(z_s|z_t)}{p_\theta(z_s|z_t)}\right] \\
&= \mathrm{KL}\big(q_\phi(z_s|z_t) \,\|\, q^\star_\phi(z_s|z_t)\big) + \sum_{i=1}^{D} \mathbb{E}_{q_\phi(z_s|z_t)}\left[\log \frac{q_\phi(z_s^i|z_t)}{p_\theta(z_s^i|z_t)}\right] \\
&= \mathrm{KL}\big(q_\phi(z_s|z_t) \,\|\, \prod_{i=1}^{D} q_\phi(z_s^i|z_t)\big) \\
&\qquad + \sum_{i=1}^{D} \mathrm{KL}\big(q_\phi(z_s^i|z_t) \,\|\, p_\theta(z_s^i|z_t)\big)
\end{aligned}$$

We can finally substitute this result in (13), to obtain the following decomposition:

$$\begin{aligned}
&\mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s|z_t) \,\|\, p_\theta(z_s|z_t)\big)\Big] \\
&= \mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s|z_t) \,\|\, \prod_{i=1}^{D} q_\phi(z_s^i|z_t)\big)\Big] \\
&\quad + \sum_{i=1}^{D} \mathbb{E}_{q_\phi(z_t)}\Big[\mathrm{KL}\big(q_\phi(z_s^i|z_t) \,\|\, p_\theta(z_s^i|z_t)\big)\Big]
\end{aligned} \tag{15}$$

A.2. Derivation of the Marginal KL Bound

In this section, we derive the relationship between the conditional KL divergence used in the standard ELBO and the marginal KL divergence that represents the ideal objective for the reverse process.

Let the conditional loss term at a given timestep be:

$$\mathcal{L}_{\text{cond}} = \mathbb{E}_{q(x)}\mathbb{E}_{q(z_t|x)}\left[D_{\mathrm{KL}}\big(q_\phi(z_s \mid z_t, x) \,\|\, p_\theta(z_s \mid z_t)\big)\right] \tag{16}$$

Using the identity $q(x, z_t) = q(z_t \mid x)q(x) = q(x \mid z_t)q(z_t)$, we can rewrite the objective as:

$$\mathcal{L}_{\text{cond}} = \mathbb{E}_{q(z_t)}\left[\mathbb{E}_{q(x|z_t)}\left[D_{\mathrm{KL}}\big(q_\phi(z_s \mid z_t, x) \,\|\, p_\theta(z_s \mid z_t)\big)\right]\right] \tag{17}$$

Focusing on the inner expectation, we observe that the KL divergence is convex with respect to its first argument. By applying Jensen's Inequality, we have:

$$\begin{aligned}
&\mathbb{E}_{q(x|z_t)}\left[D_{\mathrm{KL}}\big(q_\phi(z_s \mid z_t, x) \,\|\, p_\theta(z_s \mid z_t)\big)\right] \\
&\quad \geq D_{\mathrm{KL}}\left(\mathbb{E}_{q(x|z_t)}[q_\phi(z_s \mid z_t, x)] \,\middle\|\, p_\theta(z_s \mid z_t)\right)
\end{aligned} \tag{18}$$

The term $\mathbb{E}_{q(x|z_t)}[q_\phi(z_s \mid z_t, x)]$ is exactly the definition of the aggregated posterior $q_\phi(z_s \mid z_t)$. Therefore:

$$\mathcal{L}_{\text{cond}} \geq \mathbb{E}_{q(z_t)}\left[D_{\mathrm{KL}}\big(q_\phi(z_s \mid z_t) \,\|\, p_\theta(z_s \mid z_t)\big)\right] \tag{19}$$

Notice that we can also go further and exactly characterize the gap between the two terms. Using the definition of the aggregated posterior $q_\phi(z_s \mid z_t) = \mathbb{E}_{q(x|z_t)}[q_\phi(z_s \mid z_t, x)]$, we obtain the exact identity

$$\begin{aligned}
\mathcal{L}_{\text{cond}} &= \mathbb{E}_{q(z_t)}\left[\mathrm{KL}\big(q_\phi(z_s \mid z_t) \| p_\theta(z_s \mid z_t)\big)\right] \\
&\quad + \mathbb{E}_{q(z_t)}\mathbb{E}_{q(x|z_t)}\left[\mathrm{KL}\big(q_\phi(z_s \mid z_t, x) \| q_\phi(z_s \mid z_t)\big)\right].
\end{aligned} \tag{20}$$

The second term corresponds to the conditional mutual information $I(z_s; x \mid z_t)$ and quantifies the gap between the tractable ELBO and the ideal marginal objective.

A.3. Equivalence of REINFORCE estimator expression

Assuming that the K-L divergence $\mathrm{KL}(z_t, x)$ is indeed independent of ϕ, the gradient inside the expectation of (13) simplifies:

$$\begin{aligned}
\nabla_\phi\Big(\frac{q_\phi(z_t|x)}{[\,q_\phi\,]_{\text{sg}}}\,\mathrm{KL}(z_t, x)\Big) &= \mathrm{KL}(z_t, x)\,\nabla_\phi\Big(\frac{q_\phi(z_t|x)}{[\,q_\phi\,]_{\text{sg}}}\Big) \\
&= \mathrm{KL}(z_t, x)\,\frac{1}{[\,q_\phi\,]_{\text{sg}}}\,\nabla_\phi q_\phi(z_t|x) \\
&= \mathrm{KL}(z_t, x)\,\frac{q_\phi(z_t|x)}{[\,q_\phi\,]_{\text{sg}}}\,\nabla_\phi \log q_\phi(z_t|x).
\end{aligned}$$

Inserting this back into (13) gives

$$\hat{g}_\phi = \mathbb{E}_{u(t)q_\phi(z_t|x)}\Big[\frac{q_\phi(z_t|x)}{[\,q_\phi\,]_{\text{sg}}}\,\mathrm{KL}(z_t, x)\,\nabla_\phi \log q_\phi(z_t|x)\Big]. \tag{21}$$

Recognise that the scaling factor $\frac{q_\phi}{[\,q_\phi\,]_{\text{sg}}}$ equals 1, the estimator reduces to

$$\hat{g}^{\text{REIN}}_\phi = \mathbb{E}_{u(t)q_\phi(z_t|x)}\big[\mathrm{KL}(z_t, x)\,\nabla_\phi \log q_\phi(z_t|x)\big],$$

which is precisely the standard REINFORCE (score-function) estimator for $\nabla_\phi \mathbb{E}_{q_\phi}[\mathrm{KL}(z_t, x)]$. Thus, under the assumption that $\mathrm{KL}(z_t, x)$ does not depend on ϕ, equation (13) is equivalent to the standard REINFORCE estimator. The stop-gradient merely prevents the denominator from contributing to the gradient by treating it as a constant in backpropagation.

A.4. Derivation of optimal REINFORCE baseline

Consider the baseline $b(x, t)$ (independent of z_t). Then

$$g^{\text{REIN}}_\phi(z_t) = \frac{q_\phi(z_t|x)}{[\,q_\phi\,]_{\text{sg}}}\,\mathrm{KL}(z_t, x)\,\nabla_\phi \log q_\phi(z_t|x).$$

A baseline can be introduced without bias because $\mathbb{E}_{q_\phi}[\nabla_\phi \log q_\phi] = 0$:

$$\tilde{g}^{\text{REIN}}_\phi(z_t) = \frac{q_\phi(z_t|x)}{[\,q_\phi\,]_{\text{sg}}}\,\big(\mathrm{KL}(z_t, x) - b(x, t)\big)\,\nabla_\phi \log q_\phi(z_t|x).$$

We seek $b^*(x,t)$ that minimises the variance of $\tilde{g}_\phi^{\text{REIN}}$. Since the expectation is unchanged, it suffices to minimise the second moment

$$\mathbb{E}_{q_\phi(z_t|x)}\left[\left\|\big(\mathrm{KL}(z_t,x)-b\big)\ \nabla_\phi \log q_\phi(z_t|x)\right\|^2\right].$$

Define the weight

$$w(z_t)=\left\|\nabla_\phi \log q_\phi(z_t|x)\right\|^2.$$

Then the second moment is

$$\mathbb{E}_{q_\phi(z_t|x)}\big[w(z_t)\,(\mathrm{KL}(z_t,x)-b)^2\big].$$

Minimising this weighted least-squares expression gives

$$b^*(x,t)=\frac{\mathbb{E}_{q_\phi(z_t|x)}\big[w(z_t)\,\mathrm{KL}(z_t,x)\big]}{\mathbb{E}_{q_\phi(z_t|x)}\big[w(z_t)\big]}.$$

Substituting back $w(z_t)$,

$$b^*(x,t)=\frac{\mathbb{E}_{q_\phi(z_t|x)}\Big[\|\nabla_\phi \log q_\phi(z_t|x)\|^2\ \mathrm{KL}(z_t,x)\Big]}{\mathbb{E}_{q_\phi(z_t|x)}\Big[\|\nabla_\phi \log q_\phi(z_t|x)\|^2\Big]}. \tag{22}$$

A.5. Sinkhorn Algorithm

The solution given by the Sinkhorn algorithm is

$$P^*=\mathrm{diag}(u)K_\varepsilon \mathrm{diag}(v)$$

where $K_\varepsilon=\exp(-C/\varepsilon)$ and u,v are scaling vectors that can be found by iterating

$$u_r \leftarrow \frac{u_{s,r}}{(K_\varepsilon v)_r} \quad \text{and} \quad v_c \leftarrow \frac{u_{t,r}}{(K_\varepsilon^\top u)_c} \tag{23}$$

Finally, this defines the conditional probabilities

$$q_\phi(z_s^i=r|z_t^i=c,x)=\frac{P^*_{r,c}}{u_{t,c}} \tag{24}$$

A.6. Full results of experiments

MNIST The following three tables show the mean of the lowest FID score achieved during training on MNIST, evaluated every 5 epochs and averaged over multiple runs. We report results for all combinations of TC weight, baseline decay rate, and coupling method.

Table 4 shows results using maximum coupling, which corresponds to the original FLDD method when TC weight is zero and no baseline is used. Several patterns emerge. First, the variance-reduction baseline provides consistent improvements when combined with moderate TC regularization: the best maximum coupling result (31.55) is achieved with TC weight 10^{-4} and $\beta=0.99$, representing a 23% improvement over the FLDD baseline. Second, TC regularization exhibits a clear sweet spot: weights in the range 10^{-4} to 10^{-2} yield improvements, while larger weights ($\geq 10^{-1}$) significantly degrade performance. This suggests that excessive TC penalization overly constrains the forward process, preventing it from learning useful structure. Third, the interaction between baseline and TC weight is non-trivial; for instance, at TC weight 10^{-3}, $\beta=0.9$ achieves 32.73 while $\beta=0.99$ yields 51.26, indicating sensitivity to hyperparameter choices.

Table 4. FID scores for Maximum Coupling method on MNIST with 4 timesteps, mean over 3 runs.

TC Weight	Baseline	FID ↓
0	None	40.80
0	$\beta=0.9$	46.41
0	$\beta=0.99$	44.93
10^{-4}	None	42.52
10^{-4}	$\beta=0.9$	39.20
10^{-4}	$\beta=0.99$	31.55
10^{-3}	None	46.17
10^{-3}	$\beta=0.9$	32.73
10^{-3}	$\beta=0.99$	51.26
10^{-2}	None	47.94
10^{-2}	$\beta=0.9$	33.25
10^{-2}	$\beta=0.99$	49.54
10^{-1}	None	65.20
10^{-1}	$\beta=0.9$	63.33
10^{-1}	$\beta=0.99$	60.40
10^{0}	None	64.56
10^{0}	$\beta=0.9$	63.72
10^{0}	$\beta=0.99$	64.19
10^{1}	None	65.46
10^{1}	$\beta=0.9$	69.20
10^{1}	$\beta=0.99$	65.37

Table 5 presents results using entropic optimal transport with a fixed cost matrix $C_{u,v}=\mathbf{1}_{u\neq v}$. Compared to maximum coupling, the Sinkhorn-based coupling already improves performance even without TC regularization or baseline: the vanilla Sinkhorn result (35.23) outperforms the FLDD baseline (40.80) by 14%. This suggests that the smoother transport plans produced by entropic OT provide better gradient signal during training. The best result with fixed cost matrix (27.88) is achieved at TC weight 10^{-3} without baseline, though comparable performance is obtained with $\beta=0.9$. Interestingly, for this coupling method, the baseline appears less critical than for maximum coupling, possibly because the Sinkhorn coupling already provides sufficient gradient smoothness.

Table 6 shows results with a learnable cost matrix, which

Table 5. FID scores for Sinkhorn Algorithm on MNIST with 4 timesteps, mean over 3 runs.

TC Weight	Baseline	FID ↓
0	None	35.23
0	$\beta = 0.9$	28.05
0	$\beta = 0.99$	30.57
10^{-4}	None	52.39
10^{-4}	$\beta = 0.9$	31.09
10^{-4}	$\beta = 0.99$	42.16
10^{-3}	None	27.88
10^{-3}	$\beta = 0.9$	27.92
10^{-3}	$\beta = 0.99$	29.91
10^{-2}	None	42.49
10^{-2}	$\beta = 0.9$	43.73
10^{-2}	$\beta = 0.99$	37.1
10^{-1}	None	62.33
10^{-1}	$\beta = 0.9$	67.84
10^{-1}	$\beta = 0.99$	59.34
10^{0}	None	68.29
10^{0}	$\beta = 0.9$	64.57
10^{0}	$\beta = 0.99$	59.16
10^{1}	None	65.97
10^{1}	$\beta = 0.9$	64.73
10^{1}	$\beta = 0.99$	61.02

Table 6. FID scores for Sinkhorn with Learnable Cost Matrix on MNIST with 4 timesteps, mean over 2 runs.

TC Weight	Baseline	FID ↓
0	None	62.68
0	$\beta = 0.9$	27.83
0	$\beta = 0.99$	29.92
10^{-4}	None	29.85
10^{-4}	$\beta = 0.9$	27.45
10^{-4}	$\beta = 0.99$	26.68
10^{-3}	None	39.60
10^{-3}	$\beta = 0.9$	23.32
10^{-3}	$\beta = 0.99$	28.58
10^{-2}	None	27.04
10^{-2}	$\beta = 0.9$	44.38
10^{-2}	$\beta = 0.99$	29.09
10^{-1}	None	70.42
10^{-1}	$\beta = 0.9$	60.94
10^{-1}	$\beta = 0.99$	64.94
10^{0}	None	69.40
10^{0}	$\beta = 0.9$	68.36
10^{0}	$\beta = 0.99$	66.52
10^{1}	None	67.57
10^{1}	$\beta = 0.9$	77.50
10^{1}	$\beta = 0.99$	66.67

achieves our best overall result of 22.31 FID at TC weight 10^{-3} with $\beta = 0.9$. This represents a 45% improvement over the FLDD baseline. The learnable cost matrix allows the model to adapt the transport cost structure to the data, which appears particularly beneficial when combined with moderate TC regularization. However, we observe that training with a learnable cost matrix without baseline ($\beta =$ None) can be unstable, as evidenced by the poor result at TC weight 0 (62.68). The baseline appears to stabilize optimization when the cost matrix is also being learned, likely by reducing variance in the gradient estimates that now must account for both the forward process and cost matrix parameters.

Summary of MNIST results Across all experiments, we observe that: (1) each proposed contribution (TC regularization, OT coupling, learnable cost matrix, variance-reduction baseline) provides improvements, (2) the contributions are largely complementary, with the best results achieved by combining them, (3) TC regularization is most effective at small weights (10^{-4} to 10^{-3}), with larger weights degrading performance, and (4) the variance-reduction baseline is particularly important when training with a learnable cost matrix.

Mixture of two Gaussians The following line graph shows the Maximum Mean Discrepancy (MMD) between the real mixture-of-two-gaussians dataset and 5000 of the model's samples, such as seen in Figure 4 at $T = 0$, averaged over 6 experiments for each TC weight. The dataset is a discretized version of the mixture $0.6 \cdot \mathcal{N}(\mu = (15, 15), \sigma = 5I) + 0.4 \cdot \mathcal{N}(\mu = (25, 35), \sigma = 10I)$ constrained to $\{0, \ldots, 49\}^2$. We used Maximum Coupling for these experiments.

This dataset provides an interesting contrast to MNIST: here the discrete alphabet is large ($|\mathcal{Z}| = 50$) but the dimensionality is small ($D = 2$), whereas MNIST has a binary alphabet but high dimensionality ($D = 784$). The results show that TC regularization is beneficial across a wider range of weights on this task, with performance improving monotonically as TC weight increases up to 10. This difference may be explained by the nature of the factorization mismatch: with only two dimensions, the total correlation is limited to pairwise dependencies between the x and y coordinates, making it easier for the TC penalty to effectively encourage independence.

In experiments, the baseline FLDD model often generated vectors near $(15, 35)$ and $(25, 15)$, i.e., it would sample the x-coordinate from one Gaussian mode and the y-coordinate from the other, even though the baseline model is technically capable of learning the correct distribution according to (Anonymous, 2025). This failure mode directly illustrates the consequence of the factorization assumption: when the

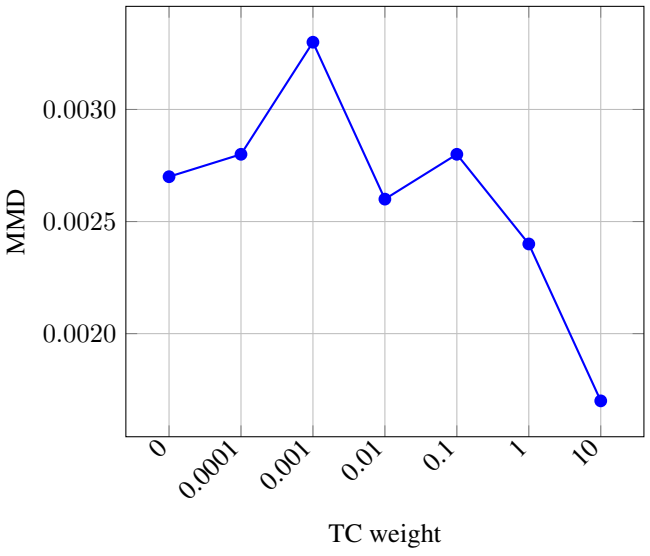

Figure 5. MMD scores on the mixture-of-two-gaussians dataset for different TC weights. Lower is better.

reverse process independently predicts each coordinate, it may fail to capture the correlation structure that links them. Introducing TC-regularization significantly reduced the occurrences of these ”mixed” samples in our experiments, pushing the model towards the correct joint distribution, as quantified by Figure 5. This provides qualitative confirmation that our TC penalty successfully encourages the forward process to induce targets that are more amenable to factorized modeling.